

Notes: Murfin and Njoroge (RFS, 2015)
The Implicit Costs of Trade Credit Borrowing by Large Firms

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1 Big picture

Murfin and Njoroge (2015) study a form of trade credit that looks puzzling from the standard financial-constraint view: very large, investment-grade retail buyers borrow from much smaller suppliers by paying them slowly. The paper asks whether these delayed payments matter for suppliers' real decisions, especially investment.

The central empirical finding is that slow payment by large buyers is costly for suppliers. When important retail buyers increase payable days, suppliers' receivables increase and suppliers cut capital expenditures and other spending. The effect is strongest when supplier financial constraints are likely to bind: for unrated suppliers, during periods of tight bank credit, and after the failure/distress of CIT, a major lender to many small and medium-sized firms.

The economic mechanism is simple. A large buyer's decision to pay later creates a working-capital financing need at the supplier. If the supplier could cheaply borrow from banks or capital markets, this would not matter much. But if the supplier is financially constrained, the supplier finances the buyer partly by using cash and cutting investment. Thus trade credit can shift financing burdens from low-cost-of-capital buyers to high-cost-of-capital suppliers.

2 Incremental contribution of the paper

1. **Documents the reverse trade-credit relationship.** The classic trade-credit literature emphasizes small constrained buyers borrowing from larger suppliers. This paper focuses on the opposite: large, highly rated buyers extracting credit from smaller suppliers.
2. **Links trade-credit terms to real supplier investment.** The paper moves beyond describing payment terms and shows that longer buyer payable days are associated with lower supplier capital expenditure and SG&A spending.
3. **Uses matched buyer-supplier relationships.** The hand-collected sample links 40 large investment-grade retailers to 723 smaller suppliers, producing 1,063 unique buyer-supplier pairs. This allows the authors to study how a buyer's aggregate payment policy affects its linked suppliers.
4. **Separates financing-friction effects from generic buyer-supplier correlations.** The paper shows that investment sensitivity to payment speed is concentrated among suppliers and periods where external finance is costly: unrated suppliers, tight bank credit periods, and suppliers linked to CIT during its distress.
5. **Provides quasi-experimental evidence using CIT distress.** The collapse and lending contraction of CIT is used as a shock to supplier access to finance. Suppliers connected to CIT became more sensitive to buyer payment terms after 2007 relative to comparable suppliers linked to the same buyers.
6. **Shows that forgone investment may be costly.** Longer buyer payment delays are followed by lower long-run operating profitability for constrained suppliers, suggesting that the investment cuts are not merely reductions in wasteful spending.

3 Research question and economic logic

The paper’s core question is:

When large investment-grade buyers pay suppliers more slowly, do suppliers merely rearrange financing, or do they cut real spending and investment?

Under a Modigliani-Miller style irrelevance benchmark, the identity of the party financing inventory should not matter. If a buyer delays payment, the supplier could borrow externally and continue investing. But with financing frictions, the supplier may be unable to replace the cash cheaply. The opportunity cost of extending trade credit is then positive and increasing in the supplier’s degree of financial constraint.

The paper’s interpretation is a **crowding-out mechanism**: slower buyer payment increases supplier accounts receivable, which absorbs cash. If external financing is costly, the supplier responds by reducing investment, cutting cash holdings, or both.

4 Institutional setting: trade credit by large retail buyers

Trade credit is large in aggregate. The paper notes that trade payables are the second-largest liability on the aggregate balance sheet of U.S. nonfinancial businesses. In retail, the economic role is especially clear: trade credit finances the gap between inventory acquisition and final sale to customers.

Large retailers can use payment terms as a source of working capital. The sample includes large retailers such as Walmart, Home Depot, Lowe’s, Target, Costco, Best Buy, Walgreens, Kroger, Macy’s, and others. These buyers are much larger and safer than their typical supplier. At the median, the supplier’s market capitalization is only about one-tenth of 1% of the buyer’s market capitalization, and supplier loan spreads are far higher than buyer loan spreads.

The paper’s setting is attractive because the buyers are large relative to each supplier. A single small supplier is unlikely to drive a retailer’s overall payable policy. This makes changes in buyer payable days a plausible source of variation in the supplier’s receivable pressure.

5 Data and sample construction

The sample runs from 1985 to 2009. Buyers are selected to satisfy three conditions:

- the buyer is identifiable as a major customer in Compustat customer segment data;
- the buyer is in retail, using NAICS 44/45 or GICS group 2550;
- the buyer has an investment-grade S&P long-term issuer credit rating.

Suppliers are identified from Compustat customer segment disclosures and linked to buyers using hand-cleaned buyer names. The final sample has 1,063 buyer-supplier pairs, 40 unique retail buyers, and 723 unique suppliers.

The key variables are:

- **Buyer payable days:** accounts payable divided by purchases, multiplied by 360. Purchases are cost of goods sold plus change in inventory.
- **Supplier receivable days:** trade accounts receivable divided by sales, multiplied by 360.
- **Supplier investment:** capital expenditures divided by lagged assets.
- **SG&A spending:** selling, general, and administrative expenditure divided by lagged assets.
- **Controls:** supplier cash flow, supplier q, buyer q, buyer-supplier fixed effects, and year fixed effects.

6 Empirical strategy

The baseline specification studies whether supplier investment changes when an important buyer pays more slowly:

$$\frac{I_{i,t}}{A_{i,t-1}} = \alpha_{i,j} + \eta_t + \beta_1 \text{BuyerPayableDays}_{j,t} + \beta_2 \frac{CF_{i,t}}{A_{i,t-1}} + \beta_3 Q_{i,t-1} + \varepsilon_{i,t}.$$

The buyer-supplier fixed effect $\alpha_{i,j}$ absorbs time-invariant features of the match. Year fixed effects absorb common macro trends. The coefficient of interest is β_1 : if it is negative, then slower payment by the buyer is associated with lower supplier investment.

The identifying idea is that the buyer’s aggregate payable policy is not determined by any one small supplier. The supplier takes the buyer’s payment policy as given, but the policy can have a large impact on the supplier because the buyer may account for a large share of the supplier’s sales.

A key example is Home Depot. In 2001, Home Depot changed policy and began paying suppliers more slowly. Lowe’s, a close comparable buyer, did not make the same change at the same time. This event motivates a difference-in-differences comparison between Home Depot suppliers and Lowe’s suppliers.

7 Main findings

7.1 Buyer payment delays increase supplier receivable pressure

The first result is that buyer payable days pass through into supplier receivable days. A one-month increase in buyer payable days is associated with roughly a 0.184 month increase in supplier receivable days. This confirms that the buyer payment measure captures a real trade credit demand placed on suppliers.

7.2 Slower payment reduces supplier spending

The key investment result is negative and economically meaningful. A one-month delay in payment by a significant buyer is associated with a reduction in supplier capital expenditures of about 1.2% of lagged assets. The same delay is associated with a reduction in SG&A of about 1.5% of lagged assets.

The paper emphasizes that these magnitudes are economically important because the typical supplier's sales to a major buyer can be large relative to the supplier's assets. Even a modest extension of payment terms can therefore generate a large working-capital shortfall.

7.3 Suppliers do not simply issue new debt

The financing-margin tests show that supplier receivable growth is not financed mainly with new debt. The coefficient of buyer payable days on debt issuance is statistically insignificant. Suppliers do reduce cash holdings, but the cash response only partially fills the funding gap. The remaining adjustment comes from reduced spending.

7.4 The effect is strongest when financial constraints bind

The paper presents three complementary tests of financial constraints.

First, rated suppliers are less sensitive to buyer payment delays. For unrated suppliers, buyer payable days reduce capital expenditure. The interaction with a ratings dummy offsets much of this effect. Rated suppliers can also issue debt in response to slow buyer payment.

Second, sensitivity is stronger when bank credit is tight. In periods when more senior loan officers report tightening lending standards, a one-month buyer payment delay has a much larger negative effect on supplier investment.

Third, suppliers linked to CIT became more sensitive to buyer payment speed after CIT's distress in 2007. Since CIT's failure reduced credit access for its relationship borrowers, this finding supports the view that the investment response is driven by financing constraints.

7.5 Payment delays predict lower long-run profitability

The paper asks whether investment cuts represent inefficient underinvestment or simply reduced overinvestment. The evidence suggests the former. For affected suppliers, increases in buyer payable days predict lower operating profitability several years later, especially four to five years ahead. This pattern is consistent with missed investment opportunities, maintenance deferral, or other costly reductions in productive spending.

8 Interpretation and takeaways

The paper's main message is that trade credit is not only a benign contract term. When strong buyers delay payment to weaker suppliers, financing is shifted from low-friction firms to high-friction firms. This can reduce investment at the supplier level and may destroy value.

The results also help reinterpret buyer market power. Slow payment terms may be a way for large retailers to improve working capital at the expense of suppliers. Even if the buyer's own financing costs fall, the total supply-chain financing cost can rise if the supplier's financing frictions are more severe.

The paper does not prove that slow payment is always inefficient. Trade credit may have benefits related to product quality, relationship contracting, bargaining, or inventory management. But the

evidence shows that the opportunity cost of supplier financing can be substantial, especially during credit contractions.

9 Appendix materials in the paper

Appendix A gives examples of 10-K disclosures that make the paper’s mechanism concrete. Paradise Inc. reports major receivable concentration with Walmart. Universal Forest Products explicitly states that operating cash flow fell partly because it extended Home Depot payment terms by an additional 15 days. Home Depot itself reports that days payable outstanding increased from 23 days to 34 days around its 2001 working-capital policy change.

Appendix B lists the 40 major retail buyers in the sample. The list includes large general merchandisers, department stores, drugstores, grocers, home-improvement retailers, office-supply firms, electronics retailers, apparel retailers, and other major retail chains. This broad buyer list helps show that the mechanism is not driven by a single retailer.

10 Extracted figures and tables from the paper

10.1 Table 1: Summary Statistics

Read. Panel A uses the full Compustat sample to show that the smallest firms are large net trade-credit borrowers, but the largest two size deciles are also net trade-credit borrowers. Panel B describes the matched buyer-supplier sample. Buyers are huge relative to suppliers: the median buyer market capitalization is about \$22.7 billion, while the median supplier market capitalization is about \$29 million. Buyers have far cheaper debt financing: median buyer all-in loan spreads are around 0.35%, compared with 2.25% for suppliers. Buyers are usually investment grade, whereas the median supplier is unrated.

Interpretation. This table establishes the central puzzle. If trade credit were only about constrained small buyers borrowing from unconstrained large sellers, the largest firms should not be net borrowers from smaller firms. Yet large retailers borrow heavily from small suppliers. The financing wedge is also large: the party providing credit is usually much smaller and faces a higher cost of external capital. This makes it plausible that large-buyer trade credit can destroy value at the supplier level.

Identification takeaway. The supplier is tiny relative to the buyer, but the buyer is important to the supplier. This asymmetry is the foundation for the empirical strategy: the supplier takes the buyer’s payment policy as given, but the policy can materially move the supplier’s balance sheet.

10.2 Figure 1: Payable Days Examples

Read. Figure 1 plots payable days for Home Depot and Lowe’s. Home Depot sharply increased payable days after the 2001 policy change, moving from around the mid-20s to roughly 40 days. Lowe’s shows a later increase, especially after 2005.

Interpretation. The figure shows that buyer payment policy changes discretely, not only smoothly with macro conditions. Home Depot’s jump is especially useful because it is tied to a stated working-

capital policy: management recognized that fast payment made Home Depot a financing source for its suppliers and changed the policy.

Identification takeaway. The staggered timing helps the paper compare suppliers exposed to a buyer that changes terms with suppliers exposed to buyers that have not yet changed terms.

Table 1
Summary statistics

Panel A: Trade credit usage by firm size

Size decile (assets)	Smallest 1	2	3	4	5	6	7	8	9
Median net trade credit days (payable days - receivable days)	61.51	6.60	0.00	-6.84	-3.07	-2.75	-2.13	-1.55	1.17

Panel B: Sample description

	(1) Relationship length (years)	(2) Buyer payable days	(3) Supplier receivable days	(4) Buyer Mt. Cap. (millions)	(5) Supplier Mt. Cap. (millions)	(6) Supplier-buyer sales/buyer COGS (%)	(7) Supplier-buyer sales/supplier Sales (%)	(8) Supplier CAPX/lagged assets	(9) Supplier SG&A/lagged assets	(10) Buyer all-in-lian spread (DuelScan)	(11) Supplier all-in-lian spread (DuelScan)
Mean	7.79	39.37	56.95	44,276	352	0.16	25.12	0.06	0.93	0.55%	2.23%
SD	4.92	16.97	60.84	48,778	1291	0.21	18.36	0.56	3.51	0.65%	1.43%
10%	3	25.98	24.63	3,555	3	0.00	7.65	0.01	0.16	0.17%	0.45%
25%	4	31.09	35.79	6,487	9	0.02	10.96	0.02	0.25	0.25%	1.25%
Median	7	35.21	50.79	22,659	29	0.07	14.95	0.03	0.37	0.35%	2.25%
75%	10	41.17	68.27	87,485	141	0.20	23.00	0.06	0.59	0.55%	3.00%
90%	16	60.97	90.93	116,340	641	0.48	38.00	0.10	0.88	1.12%	4.00%

Using the full Compustat sample (excluding financials) as of 2009, median net trade credit days, defined as payable days minus receivable days, are presented for ten size by total assets. The sample period is from 1985 to 2009 and consists of 1,060 unique buyer-seller pairs involving forty unique retail buyers and 723 unique sellers. The various summary statistics from our sample are reported below. Payable days for buyers are calculated as accounts payable (item AP) divided by purchases (cost of goods inventory) and multiplied by 360. Receivable days for sellers are calculated as trade accounts receivable (item RECTR or RECT if RECTR is missing) divided by sales s . With the assumption of 360 days in a year, this distribution size reported for annual observations occurring on the first of a bi-monthly or quarterly date. Further details are reported in the appendix.

Table 1: Summary Statistics

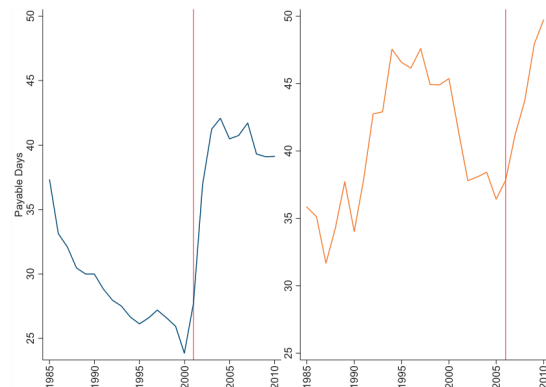


Figure 1
Payable days examples
Figure 1 plots the payable days for two major retailers in the sample. In 2001, Home Depot (left) announced it would pay its suppliers roughly 15 days more slowly than it had previously. In 2006, Lowe's (right) began

Figure 1: Payable Days Examples

10.3 Figure 2: Payable Days Examples (Continued)

Read. Figure 2 plots the median payable days for the paper's sample retailers and for the largest Compustat retailers. Both series trend upward from the late 1980s to 2010, moving from around 28-31 days to around 37-40 days.

Interpretation. The figure shows that slower payment is not an isolated Home Depot event. The entire retail sector shifted toward longer payable days. The trend is economically important because a few extra payable days can represent a large working-capital shock for small suppliers with high sales relative to assets.

Identification takeaway. The upward trend creates a need for year fixed effects. The useful variation comes from the fact that individual retailers adjust at different times.

10.4 Table 2: Small Sample Evidence on the Effects of Trade Terms

Read. Table 2 implements a Home Depot versus Lowe's difference-in-differences test around Home Depot's 2001 payment-policy change. Home Depot suppliers reduce capital expenditures relative to Lowe's suppliers after 2001. The interaction coefficient is around -0.025 to -0.032, depending on specification.

Interpretation. A decline of roughly 2.5% of lagged assets is large for capital expenditure. The result provides a transparent event-study style illustration of the broader mechanism: when a major buyer delays payment, suppliers cut investment.

Identification takeaway. This table is not the full identification strategy, but it makes the mechanism visually and statistically credible. It also helps rule out a simple buyer-distress story because both Home Depot and Lowe's were large, healthy investment-grade buyers at the time.

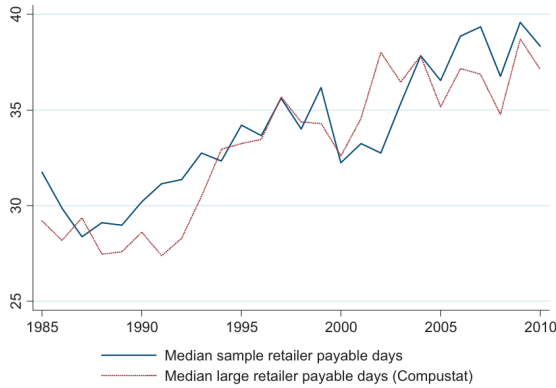


Figure 2
Payable days examples (continued)
Below, we plot the median payable days by fiscal year for the retailers in our sample, alongside the median payable days for the top size decile (by assets) of Compustat retailers, defined based on GIC group classification (2550) and with size deciles recalculated each year.

Figure 2: Payable Days Examples (Continued)

Table 2
Small sample evidence on the effects of trade terms

	(1) Capital expenditure	(2) Capital expenditure	(3) Capital expenditure
Home Depot supplier	0.030* (0.017)	0.015 (0.011)	0.017 (0.011)
Home Depot supplier X post-2001	-0.032* (0.016)	-0.025* (0.014)	-0.026** (0.013)
Post-2001	0.003 (0.014)		
Cash flow	0.081 (0.049)	-0.015 (0.064)	-0.014 (0.065)
q	0.014** (0.007)	0.015* (0.008)	0.015* (0.008)
Buyer q			-0.001 (0.003)
Supplier fixed effects	no	yes	yes
Year fixed effects	no	yes	yes
Observations	281	281	281
R ²	0.133	0.152	0.152

Columns 1—3 report the results from a difference-in-differences regressions. We compare the investment activity of Home Depot suppliers to that of Lowe's matched suppliers around Home Depot's 2001 decision to delay payment terms by 15 days going forward for its vendors. We use a symmetric window of observation between 1997 and 2004. Columns 2 and 3 add supplier and year fixed-effects. Standard errors are clustered at the supplier level. See Figure A1 for an analysis of parallel trends in the investment activity of treatment and control groups. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table 2: Small Sample Evidence on the Effects of Trade Terms

10.5 Table 3: Buyer Payable Days and Supplier Receivable Days

Read. Table 3 shows that a one-month increase in buyer payable days is associated with about a 0.184 month increase in supplier receivable days. The relationship is statistically significant after controlling for buyer-supplier fixed effects and year fixed effects. The table also shows that longer buyer payable days are associated with higher supplier markups.

Interpretation. This table verifies the first stage: buyer payable policy affects supplier receivables. The result is not merely an accounting artifact; it shows that the buyer's working-capital policy is transmitted to the supplier's balance sheet. The markup result suggests suppliers may receive some price compensation for slower payment, but the rest of the paper shows that this compensation does not fully eliminate real costs for constrained suppliers.

Identification takeaway. Without this first stage, the paper's later investment results would be hard to interpret. Table 3 confirms that buyer payable days are a meaningful measure of the financing demand imposed on suppliers.

10.6 Table 4: Working Capital and Investment

Read. Table 4 is the core investment result. A one-month increase in buyer payable days is associated with a -0.012 change in supplier capital expenditure scaled by lagged assets, significant at the 1% level. The coefficient on SG&A is -0.015 and significant at the 5% level.

Interpretation. Suppliers cut both physical investment and broader operating expenditures when buyers pay more slowly. The capital-expenditure effect is especially large relative to mean supplier capex of about 6% of assets. This implies that payment terms can affect real production capacity, not just short-term accounting balances.

Economic magnitude. A one-month payment delay can create a financing gap of roughly 9% of lagged assets for a supplier whose sales to the buyer are large relative to assets. A 1.2% capex reduction means that a meaningful part of the financing gap is absorbed by cutting investment.

Table 3
Buyer payable days and supplier receivable days

	(1) Receivable days (in months)	(2) Receivable days (in months)	(3) Supplier markup
Buyer payable days (in months)	0.185** (0.088)	0.184*** (0.067)	
ln(Buyer payable days)			0.061* (0.032)
Buyer q		0.005 (0.021)	
Cash flow		-0.044 (0.119)	
q		0.007 (0.039)	
Supplier X buyer fixed effects	yes	yes	yes
Year fixed effects	yes	yes	yes
Observations	5,352	4,280	5,354
R ²	0.03	0.04	0.001

Columns 1 and 2 estimate the effect of average payable days outstanding of large retailers on their suppliers' receivables. The model is estimated assuming buyer-supplier fixed effects.

$$\text{SupplierReceivableDays}_{i,t} = \alpha_i + \alpha_{ij} + \beta_1 \text{BuyerPayableDays}_{j,t} + \beta_{2-k} X_{i,j,t} + \varepsilon_{i,j,t},$$

where subscript i indexes suppliers and subscript j indexes buyers. *Buyer Payable Days* (reported in months) are calculated as the accounts payable of the buyer divided by cost of goods sold, times 30, whereas *Supplier Receivable Days* are calculated as trade receivables divided by net sales, times 30. Cash flow is operating income before depreciation divided by last period's assets, and q is calculated as the market value of equity plus book value of assets, less shareholders equity and deferred taxes, divided by assets. q is measured at the beginning of each period for the buyer and the supplier. In addition to the controls reported, regressions include buyer-supplier pair fixed effects and time fixed effects. Column (3) reports the relation between the supplier's markup, defined as (sales-cost of goods sold)/cost of goods sold, and the log of buyer payable days. Standard errors are double clustered at the level of supplier and buyer, are robust to heteroscedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table 3: Buyer Payable Days and Supplier Receivable Days

10.7 Table 5: Financing Receivable Growth

Read. Table 5 studies whether suppliers finance slower buyer payment by issuing debt or using cash. The coefficient on debt issuance is small and insignificant. The coefficient on change in cash is negative and significant at the 10% level: slower buyer payment is associated with cash drawdowns.

Interpretation. Suppliers do not appear to fill the working-capital gap with new debt. They use some cash, but the investment results show that cash is not enough. This pattern is exactly what one would expect if external financing is costly and internal liquidity is limited.

Mechanism takeaway. The result helps distinguish a financial-frictions story from a pure operating story. If suppliers could freely borrow, buyer payment delays would not require investment cuts. The absence of debt issuance points to financing constraints.

10.8 Table 6: Credit Availability and Supplier Sensitivity to Payment Delays, Part 1

Read. Table 6 interacts buyer payable days with a supplier ratings dummy. For unrated suppliers, buyer payable days significantly reduce capital expenditure. The interaction with having a rating is positive, offsetting much of the negative effect. In the debt issuance regression, the interaction is large and positive, indicating that rated firms can issue debt when buyers pay more slowly.

Interpretation. This table is one of the clearest pieces of evidence for the financial-constraint mechanism. Rated firms have better access to external capital, so they are less forced to cut investment. Unrated firms cannot easily replace the delayed cash and therefore reduce capex.

Identification takeaway. The same buyer payment variable has different effects depending on supplier financial access. This heterogeneity is difficult to reconcile with an explanation based only on buyer demand shocks or industry trends.

Table 4
Working capital and investment

	(1) Capital expenditure	(2) SG&A expenditure
Buyer payable days (in months)	-0.012*** (0.003)	-0.015** (0.007)
Buyer q	-0.002 (0.001)	-0.004 (0.005)
Cash flow	0.038*** (0.010)	0.597*** (0.028)
q	0.012*** (0.002)	0.031*** (0.006)
Supplier X buyer fixed effects	yes	yes
Year fixed effects	yes	yes
Observations	4,278	4,199
R ²	0.14	0.67

Columns 1 and 2 estimates the effect of buyer payment speed on supplier expenditures.

$$\text{SupplierInvestment}_{i,t} = \alpha_i + \alpha_{ij} + \beta_1 \text{BuyerPayableDays}_{j,t} + \beta_{2-k} X_{i,j,t} + \varepsilon_{i,j,t},$$

where subscript i indexes suppliers and subscript j indexes buyers, and investment represents capital expenditure ($\text{CAPX}/\text{AT}_{t-1}$) or alternatively selling, general, and administrative expenditures ($\text{XSGA}/\text{AT}_{t-1}$), hereafter SG&A. To avoid the mechanical impact of SG&A expenditures in operating income, Column 2 measures supplier cash flow as gross profit, defined as net sales less cost of goods sold but before SG&A. In addition to the controls reported, regressions include buyer-supplier pair fixed effects and time fixed effects. Standard errors are double clustered at the level of supplier and buyer, are robust to heteroscedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table 4: Working Capital and Investment

Table 5
Financing receivable growth

	(1) Δ Debt	(2) Δ Cash
Buyer payable days (in months)	−0.004 (0.012)	−0.012* (0.006)
Buyer q	−0.011* (0.006)	−0.000 (0.002)
Cash flow	0.165** (0.072)	0.111*** (0.042)
q	0.020*** (0.006)	0.008 (0.006)
Supplier X buyer fixed effects	yes	yes
Year fixed effects	yes	yes
Fiscal quarter dummies	yes	yes
Observations	4,283	4,282
R ²	0.04	0.04

Table 5 estimates the effect of the average payable days of buyers on net debt issuance and change in cash, each scaled by lagged assets. In addition to the controls reported, regressions include buyer-supplier pair fixed effects and time fixed effects. Standard errors are double clustered at the level of supplier and buyer, are robust to heteroscedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table 5: Financing Receivable Growth

Table 6
Credit availability and supplier sensitivity to payment delays, part 1

	(1) Capital expenditure	(2) Δ Debt
Buyer payable days (in months)	−0.012*** (0.003)	−0.008 (0.016)
Buyer payable days X ratings dummy	0.010* (0.006)	0.082*** (0.021)
Buyer q	−0.002* (0.001)	−0.016** (0.008)
Cash flow	0.037*** (0.010)	0.153** (0.072)
q	0.012*** (0.002)	0.019*** (0.006)
Supplier X buyer fixed effects	yes	yes
Other interactions	yes	yes
Year fixed effects	yes	yes
Ratings dummy	yes	yes
Observations	4,278	4,283
R ²	0.14	0.05

Table 6 estimates the interaction between whether or not a supplier was rated in the prior period and the effects of buyer payable days on supplier investment and debt issuance. In addition to the controls reported, regressions include buyer-supplier pair fixed effects and time fixed effects, a dummy for whether or not the supplier was rated, and separate interactions between the ratings dummy and supplier cash flow and q, as well as buyer q. Standard errors are double clustered at the level of supplier and buyer, are robust to heteroscedasticity.

Table 6: Credit Availability and Supplier Sensitivity to Payment Delays, Part 1

10.9 Table 7: Credit Availability and Supplier Sensitivity to Payment Delays, Part 2

Read. Table 7 interacts buyer payable days with bank-credit tightening, measured by the Federal Reserve Senior Loan Officer Survey. In loose credit periods, the uninteracted effect of buyer payable days on capex is close to zero. The interaction with credit tightening is negative and significant. The interaction in the debt regression is also negative, showing that debt financing becomes less available when credit tightens.

Interpretation. The same supplier facing the same kind of buyer payment delay is much more vulnerable when credit markets are tight. This strongly supports the view that the real effects of trade credit depend on external financing conditions.

Macro takeaway. Trade credit can amplify credit cycles. During tight credit periods, large buyers' slow payment terms impose especially high opportunity costs on suppliers and can push them to cut investment.

10.10 Table 8: Credit Availability and Supplier Sensitivity to Payment Delays, Part 3

Read. Table 8 uses CIT distress as a financing shock. The key triple interaction is Buyer Payable Days × CIT linked × post-2007. The coefficient is negative and significant across specifications. In the post-2007 period, suppliers linked to CIT cut capex more in response to buyer payment delays than comparable non-CIT suppliers.

Interpretation. This is the paper's strongest quasi-experimental evidence. CIT's lending contraction reduced credit access for relationship borrowers. If financing constraints are the mechanism, CIT-linked suppliers should become more sensitive to buyer payment terms after the shock. Table 8 shows that they do.

Identification takeaway. By comparing CIT-linked and non-CIT-linked suppliers serving the same buyers, the test helps separate supplier credit access from buyer-specific payment-policy shocks.

Table 7
Credit availability and supplier sensitivity to payment delays, part 2

	(1) Capital expenditure	(2) Δ Debt
Buyer payable days (in months)	−0.003 (0.003)	0.016 (0.019)
Buyer payable days X credit tightening (FRB Senior Loan Officer Survey)	−0.023*** (0.007)	−0.036* (0.021)
Buyer q	−0.002 (0.002)	0.001 (0.008)
Cash flow	0.063*** (0.017)	0.170* (0.102)
q	0.016*** (0.003)	0.007 (0.012)
Supplier X buyer fixed effects	yes	yes
Other interactions	yes	yes
Year fixed effects	yes	yes
Observations	3,981	3,985
R ²	0.16	0.05

Table 7 estimates the interaction between tightness in the bank market (captured by the percentage of loan officers reporting tightening credit standards in a given year) and the effects of buyer payable days on supplier investment. In addition to the controls reported, regressions include buyer-supplier pair fixed effects, time fixed effects, and separate interactions between credit tightening and supplier cash flow and q, as well as buyer q. The measure of credit tightening has been recentered at the minimum value of credit tightening (that is, the loosest period of credit, achieved during 2004), such that the coefficient on (uninteracted) buyer payable days can be interpreted as the impact of payment speed on investment and debt issuance during periods of loose credit. Standard errors are double clustered at the level of supplier and buyer, are robust to heteroscedasticity, and are

Table 7: Credit Availability and Supplier Sensitivity to Payment Delays, Part 2

Table 8
Credit availability and supplier sensitivity to payment delays, part 3

	(1) Capital expenditure	(2) Capital expenditure	(3) Capital expenditure
Buyer payable days (in months)	0.002 (0.005)	−0.003 (0.004)	−0.002 (0.003)
Buyer q	−0.000 (0.002)	−0.001 (0.001)	−0.001 (0.002)
Cash flow	0.080*** (0.012)	0.045*** (0.014)	0.046*** (0.014)
q	0.008*** (0.002)	0.009*** (0.002)	0.010*** (0.002)
CIT linked	−0.001 (0.007)		
CIT linked x post-2007	−0.004 (0.003)	−0.000 (0.007)	0.001 (0.003)
Buyer payable days x post-2007	−0.012* (0.006)	0.003 (0.002)	0.005 (0.005)
Buyer payable days x CIT linked	−0.003 (0.005)	0.008 (0.010)	0.007 (0.009)
Buyer payable days x CIT linked x post-2007	−0.016*** (0.004)	−0.010** (0.005)	−0.011** (0.005)
Supplier X buyer fixed effects	no	yes	yes
Year fixed effects/post-2007	yes	yes	yes
Ratings dummy	yes	yes	yes
Observations	1,972	1,900	1,717
R ²	0.18	0.18	0.18

Table 8 estimates the differential impact of buyer payable days on investment between suppliers linked to CIT Group and those with non-CIT lending relationships, before and after 2007. Lender relationships are identified by matching the supplier sample with DealScan prior to 2007—suppliers with no DealScan transactions are excluded from the sample. We limit the sample period to the final 10 years of data, beginning in 2000. The variable post-2007 is a dummy for any observation reported after June 2007. Of interest is the interacted coefficient of **BuyerPayableDays X CIT X post-2007**, which captures the change in CIT-linked suppliers' sensitivity to buyer payment speed following CIT's financial difficulty, benchmarked against the change in sensitivity for non-CIT linked suppliers. Interaction terms are formed after demeaning Buyer Payable Days, so the coefficients can be interpreted as the marginal effect at the mean. In addition to the controls reported, regressions include buyer-supplier pair fixed effects (Column 2 only), time fixed effects, and a dummy variable for rated suppliers. Column 3 limits the control sample of non-CIT linked firms to suppliers serving the same group of buyers as the treated firms. Standard errors are double clustered at the level of supplier and buyer, are robust to heteroscedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table 8: Credit Availability and Supplier Sensitivity to Payment Delays, Part 3

10.11 Table 9: Trade Credit, Investment, and Subsequent Performance

Read. Table 9 regresses subsequent supplier operating profit on buyer payable days. The short-horizon coefficients are not strongly negative, but the coefficients become negative and significant at longer horizons. Four- and five-year-ahead operating profits decline when buyer payable days increase.

Interpretation. This table addresses the welfare interpretation. If slower payment merely disciplined overinvestment, then lower capex should not predict lower future performance. Instead, payment delays are followed by weaker long-run profitability, suggesting that constrained suppliers cut valuable investment or productive spending.

Caution. This is not a structural welfare estimate. It does not quantify all costs and benefits of the trade-credit relationship. But it is consistent with the paper's view that slow payment can impose real deadweight costs on suppliers.

10.12 Table A1: Buyer Health and Payment Speed

Read. Table A1 examines whether buyer payable days are driven by buyer health. Buyer payable days are regressed on buyer q, changes in q, buyer profits, future profits, and CDS spreads. The coefficients are small and statistically insignificant.

Interpretation. This appendix table helps rule out a key alternative explanation: that slow payment simply proxies for buyer distress. If buyers paid slowly only when they were weakening, supplier investment cuts might reflect deteriorating demand from the buyer. The weak relationship between buyer health measures and payment speed reduces this concern.

Robustness takeaway. The table supports the interpretation that payment-policy variation is

not just a disguised buyer-fundamentals shock.

Table 9
Trade credit, investment, and subsequent performance

	(1)	(2)	(3)	(4)	(5)	(6)
	Operating profit (t)	Operating profit (t+1)	Operating profit (t+2)	Operating profit (t+3)	Operating profit (t+4)	Five-year operating profit
ln (Buyer payable days) (in months)	0.008 (0.061)	0.042 (0.070)	-0.079 (0.101)	-0.242** (0.100)	-0.284* (0.169)	-0.201** (0.091)
Buyer q	0.012 (0.021)	0.019 (0.016)	0.031 (0.029)	0.013 (0.020)	-0.007 (0.013)	0.018 (0.018)
q	0.004 (0.025)	-0.016 (0.035)	0.028 (0.047)	0.027 (0.042)	0.049 (0.035)	0.023 (0.036)
Supplier X buyer fixed effects	yes	yes	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes	yes	yes
Observations	4,288	3,912	3,584	3,180	2,779	2,757
R ²	0.03	0.02	0.03	0.03	0.04	0.04

Table 9 estimates the effect of payment speed on subsequent performance for affected suppliers. Performance is measured by operating profitability, defined as annual operating income, measured in the current period and in four leading periods (t+1 through t+4), scaled by assets in place at the beginning of period t (e.g., Operating Profit (t+3)/Operating Income (t+3)/Assets(t-1)). Each lead of operating profit is then standardized (demeaned and divided by its standard deviation) for ease of interpretation. Columns 1–5 report estimates of buyer days' effect on current and future returns to current period assets, whereas Column 6 reports the effect on aggregate 5-year operating profit. In addition to the controls reported, regressions include buyer-supplier pair fixed effects and time fixed effects. Standard errors are double clustered at the level of supplier and buyer, are robust to heteroscedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table 9: Trade Credit, Investment, and Subsequent Performance

Table A1
Buyer health and payment speed

	(1)	(2)	(3)	(4)	(5)	(6)
Buyer payable days (in months)	-0.003 (0.032)					
Buyer q _t		0.000 (0.000)				
Buyer Profits _t			-0.161 (0.498)			
ΔBuyer Profits _t				-0.126 (0.177)		
Buyer Profits _{t+1}					-0.288 (0.318)	
Buyer CDS Spreads						-0.002 (0.004)
Buyer fixed effects	yes	yes	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes	yes	yes
Observations	757	747	783	783	743	186
R ²	0.05	0.05	0.05	0.05	0.06	0.06

We report panel regressions of buyer payable days on buyer q, profits (scaled by lagged assets), and CDS spreads. CDS spreads are the monthly average spread as of the reporting date for buyers, when available, from 2001–2009. Regressions include buyer and time fixed effects. Standard errors are clustered at the level of the buyer, are robust to heteroscedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table A1: Buyer Health and Payment Speed

10.13 Table A2: Robustness

Read. Table A2 reports robustness tests. Columns 1-3 use first differences with leads and lags of buyer payable days. The contemporaneous buyer payable days coefficient remains negative and significant. The lead is near zero, which helps address reverse causality. Column 4 returns to buyer-supplier fixed effects and adds retailer subindustry-by-year fixed effects; the coefficient remains negative and significant.

Interpretation. The robustness results show that the main investment effect is not driven by fixed pair differences, simple trends, or broad retail subindustry shocks. The insignificant lead is especially useful: it suggests supplier investment does not decline before buyer payment terms change.

Identification takeaway. Table A2 strengthens the causal interpretation by showing that the timing and robustness of the effect line up with buyer payment changes rather than with pre-existing supplier investment trends.

10.14 Figure A1: Home Depot versus Lowe's

Read. Figure A1 plots capex-to-lagged-assets for Home Depot and Lowe's suppliers around Home Depot's 2001 policy change. The lines are year fixed effects from regressions controlling for supplier cash flow, supplier q, and supplier fixed effects. After the policy change, Home Depot suppliers show a sharp decline in investment relative to Lowe's suppliers.

Interpretation. This figure visually reinforces Table 2. The investment decline is not gradual and pre-existing; it opens up after Home Depot changes payment policy. The figure makes the difference-in-differences logic intuitive.

Identification takeaway. The graph supports the parallel-trends idea for the Home Depot-Lowe's comparison. It also suggests the investment cut is economically meaningful rather than a statistical artifact.

Table A2
Robustness

	First differences			
	(1)	(2)	(3)	(4)
	Capital expenditure	Capital expenditure	Capital expenditure	Capital expenditure
Buyer Payable Days (in months) _{t+1}			0.000 (0.004)	
Buyer Payable Days (in months)	-0.014*** (0.004)	-0.018*** (0.004)	-0.011*** (0.004)	-0.012*** (0.003)
Buyer Payable Days (in months) _{t-1}		-0.007 (0.006)		
Buyer q_t	-0.002* (0.001)	-0.006*** (0.002)	-0.004* (0.002)	-0.002 (0.002)
Cash flow	0.026*** (0.008)	0.036*** (0.009)	0.034*** (0.011)	0.037*** (0.011)
q	0.012*** (0.003)	0.012*** (0.003)	0.011*** (0.003)	0.013*** (0.003)
Supplier X buyer fixed effects	no	no	no	yes
Retailer NAIC (three-digit) X year fixed effects	no	no	no	yes
Year fixed effects	yes	yes	yes	yes
Observations	3,267	2,534	2,391	4,278
R^2	0.09	0.09	0.08	0.19

Columns 1–3 replace fixed effects estimation with first differences with leads and lags. Column 4 returns to buyer-supplier fixed effects and adds fixed effects for retail sub-industry X year. Using three-digit buyer NAICS, this divides the sample into eleven groups of buyers (examples include food and beverage, health and personal care, electronics and appliances, and clothing and clothing accessories) and adds year dummies for each group. Standard errors are clustered at the level of supplier and buyer, are robust to heteroscedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Table A2: Robustness

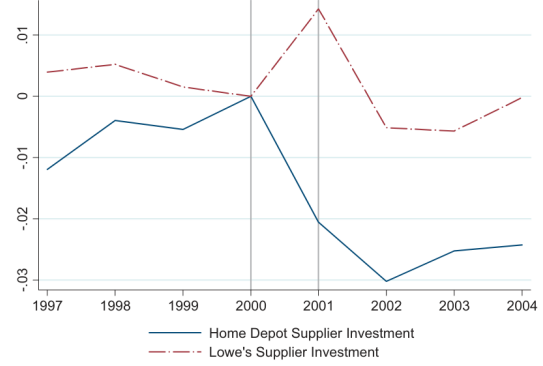


Figure A1
Home Depot versus Lowe's
Capex to lagged assets are reported for Home Depot and Lowe's suppliers around the time of 2001 policy change in which Home Depot began slower payment to suppliers. More specifically, we plot separate year fixed effects for the two supplier groups from a regression of capex/lagged assets on supplier cash flow, supplier q , and supplier fixed effects (as in Table 2, Column 2). The year 2000 dummy variable is omitted for both groups, such that the two lines represent investment deviations around the last year before the policy change, which were not attributable to cash flow, q , or firm fixed effects.

Figure A1: Home Depot versus Lowe's

10.15 Figure A2: Timing of CIT Distress

Read. Figure A2 plots the share of DealScan loans originated by CIT and CIT's share price. Both series fall sharply around the second half of 2007. The paper uses this period to define the shock to financing access for CIT-linked borrowers.

Interpretation. The graph shows that the CIT shock is sudden and severe. CIT moves from being a meaningful lender in the DealScan market to a much smaller presence, while its equity value collapses. This creates a credible shock to the financing capacity of suppliers that relied on CIT.

Identification takeaway. Figure A2 is the visual foundation for the triple-difference test in Table 8. If CIT-linked suppliers lose access to an important relationship lender, they should become more sensitive to buyer payment delays exactly after the collapse shown in the figure.

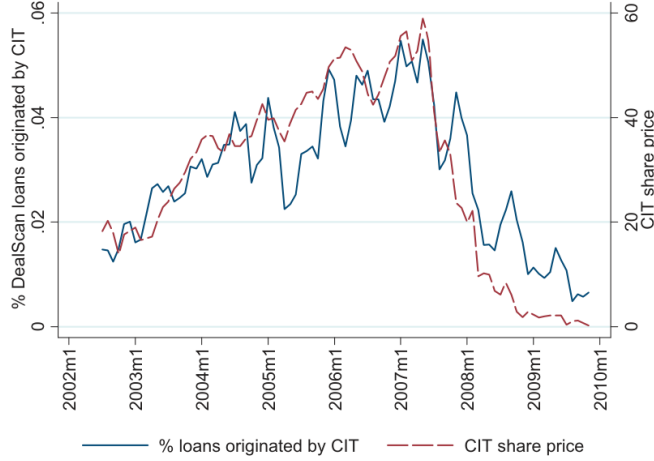


Figure A2
Timing of CIT distress

Figure A2 plots a three-month moving average of the percentage of DealScan loans, in which CIT served as a lender (as the sole bank or as part of a syndicate) on the left vertical axis. On the right vertical axis is the monthly share price of CIT equity (we plot the monthly low). We use the steep decline in lending activity and equity valuation around the second half of 2007 to define the period of relative financing constraint for CIT-linked borrowers.

Figure A2: Timing of CIT Distress

11 Bottom-line summary

The paper shows that payment terms are economically important financing contracts. Large buyers can improve working capital by paying suppliers more slowly, but this shifts financing needs to smaller suppliers that often have higher financing costs. When supplier financing constraints bind, delayed payment is associated with lower investment, lower cash, and weaker future profitability.

The main empirical contribution is the combination of matched buyer-supplier data with several identification strategies: staggered buyer payment changes, Home Depot versus Lowe's, financial-constraint heterogeneity, bank-credit tightening, and CIT distress. Across these tests, the evidence consistently supports a real cost of trade credit borrowing by large firms.